

Control of Mixed-Autonomy Traffic via Autonomous Vehicles with Lane-Changing Behavior

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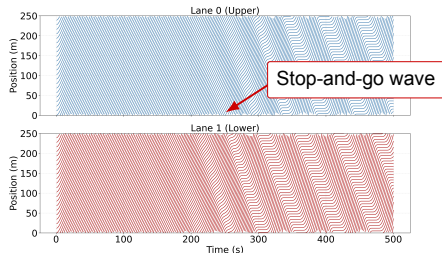
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Stop-and-go waves become a two-dimensional control problem

Stop-and-go waves are usually treated as a longitudinal-control problem.

Human lane-changing adds a second disturbance channel.

- ▶ Small braking disturbances propagate upstream.
- ▶ Lane changes abruptly modify leader–follower relationships.
- ▶ A controller confined to one lane cannot reject disturbances entering from another lane.



Lane-changing changes the active leader–follower graph.

Local longitudinal stabilization does not guarantee multi-lane system stability.

Existing methods solve different parts of the problem

Paradigm	Strength	Limitation for this problem
Rule-based control	Interpretable and deployable	Usually designed for longitudinal regulation within one lane
Optimization-based control	Handles models and constraints explicitly	Requires repeated solution of a potentially expensive optimization problem
Learning-based control	Can adapt to complex HDV interactions	Learned policies may be difficult to interpret and generalize

The missing element is an interpretable controller that explicitly coordinates traffic across lanes.

The study follows three methodological questions

- Q1.** Can one RL-controlled AV stabilize two-lane traffic when HDVs change lanes?
- Q2.** What coordination structure appears when two AVs share information?
- Q3.** Can that structure be converted into an interpretable rule-based controller?

Single-agent RL → Cooperative RL → Behavioral interpretation → PARC

A controlled testbed isolates lane-changing disturbances



- ▶ Two-lane ring road, $L = 250$ m
- ▶ $N_{\text{total}} = 44$ vehicles
- ▶ SUMO microscopic simulation
- ▶ Flow reinforcement-learning interface
- ▶ IDM longitudinal dynamics
- ▶ Heterogeneous LC2013 lane-changing behavior
- ▶ Bounded AV acceleration commands

HDVs: IDM + LC2013
AVs: learned or rule-based longitudinal control

The ring removes boundary-flow effects and isolates cross-lane interactions.

We formulate traffic control as a finite-horizon MDP

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, r, \rho_0, \gamma, H),$$

$$\max_{\pi} J(\pi) = \mathbb{E}_{\pi, \mathcal{T}} \left[\sum_{t=0}^{H-1} \gamma^t r(\mathbf{s}_t, \mathbf{a}_t, \mathbf{s}_{t+1}) \right].$$

- $\mathcal{S}$ local traffic and AV coordination states
- $\mathcal{A}$ AV acceleration and, where applicable, lane-change decisions
- $\mathcal{T}$ stochastic SUMO traffic evolution
- r traffic efficiency and coordination objective

Observation, action, and reward design determine what coordination can be learned.

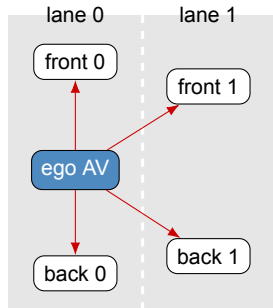
The single AV observes both lanes but acts locally

$$\mathbf{o}_t = \left[\frac{v_t^{\text{AV}}}{v_{\max}}, \mathbf{z}_t^{(0)}, \mathbf{z}_t^{(1)} \right] \in \mathbb{R}^{11},$$

$$\mathbf{z}_t^{(\ell)} = \left[\mathbf{1}_{\{\ell=\ell_t\}}, \frac{d_{f,t}^{\ell}}{L}, \frac{v_{f,t}^{\ell}}{v_{\max}}, \frac{d_{b,t}^{\ell}}{L}, \frac{v_{b,t}^{\ell}}{v_{\max}} \right],$$

$$\mathbf{a}_t = \left[\mathbf{a}_t^{\text{acc}}, \mathbf{a}_t^{\text{lc,raw}} \right],$$

r_t = instantaneous average traffic speed.

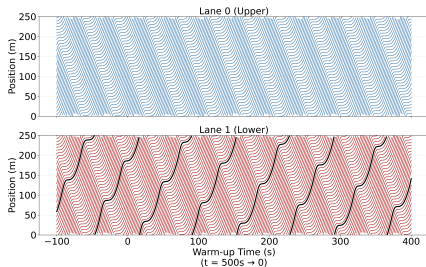


The reward is global, but the AV's physical authority remains local.

The single-agent formulation produces a locally rational policy

The AV learns to close the gap to its leader.

1. A larger gap attracts HDV lane changes.
2. A merging HDV forces braking and reduces the AV reward.
3. The AV protects its local gap—but cannot stabilize the adjacent lane.



Compression and expansion persist in both lane trajectories.

Richer observation does not compensate for insufficient control authority.

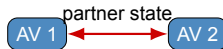
Cooperative RL changes the information and action structure

$$o_t^i = \left[\frac{v_t^i}{v_{\max}}, \frac{v_{i,t}^{\text{lead}}}{v_{\max}}, \frac{v_{i,t}^{\text{foll}}}{v_{\max}}, \frac{d_{i,t}^{\text{lead}}}{L}, \frac{d_{i,t}^{\text{foll}}}{L} \right],$$

$$o_t = \left[o_t^1, o_t^2, \frac{x_t^1 - x_t^2}{L}, \frac{v_t^1 - v_t^2}{v_{\max}} \right] \in \mathbb{R}^{12},$$

$$a_t = \begin{bmatrix} a_1(t) \\ a_2(t) \end{bmatrix} \in [-0.5, 0.5]^2.$$

- ▶ Centralized observation
- ▶ Two longitudinal control inputs
- ▶ AVs remain in assigned lanes
- ▶ Pairwise position and velocity are explicit states



The architecture adds direct authority over both lane boundaries.

The cooperative reward explicitly shapes coordination

$$\begin{aligned} r_t &= r_{\text{pos},t} + r_{\text{sync},t} + r_{\text{speed},t} + r_{\text{smooth},t}, \\ r_{\text{pos},t} &= -w_{\text{pos}} \left| \frac{x_t^1 - x_t^2}{L} \right|, \\ r_{\text{sync},t} &= -w_{\text{sync}} \left| \frac{v_t^1 - v_t^2}{v_{\text{max}}} \right|, \\ r_{\text{speed},t} &= w_{\text{speed}} \left(\frac{v_t^1 + v_t^2}{2v_{\text{max}}} \right)^2, \\ r_{\text{smooth},t} &= -w_{\text{smooth}} [(a_t^1)^2 + (a_t^2)^2]. \end{aligned}$$

Position → alignment

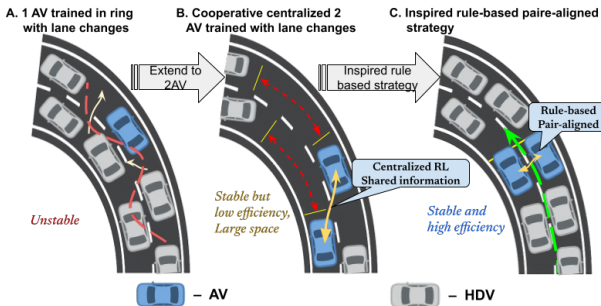
Synchronization → coherent motion

Speed → traffic efficiency

Smoothness → feasible
acceleration

Pairing is reward-shaped learned behavior, not a reward-agnostic emergent phenomenon.

The learned policy suggests a geometric control mechanism



Aligning the AVs removes the staggered gaps used by HDVs to change lanes.

1. The leading AV slows down.
2. The trailing AV closes the cross-lane offset.
3. The aligned AVs create a common moving boundary for both lanes.

The two-lane system behaves approximately as one virtual longitudinal lane.

PARC encodes the learned mechanism as a hybrid controller

$$d_p = x_p - x, \quad \Delta v_p = v_p - v,$$

$$a = \begin{cases} k_{\text{pair}} d_p, & |d_p| > d_{\text{th}}, \\ k_{\text{sync}}(v_p - v), & |v_p - v| > \delta_v, \\ k_v(v^* - v), & \text{otherwise,} \end{cases}$$

$$a \leftarrow \text{clip}(a, -a_{\text{max}}, a_{\text{max}}).$$

Formation

Regulate cross-lane position error.

Synchronization

Remove relative velocity.

Cruise

Track the shared equilibrium speed.

The controller separates geometry, coherent motion, and efficient cruise.

The cruising speed follows the equilibrium traffic geometry

$$N_\ell = \frac{N_{\text{total}}}{2} = 22,$$

$$s_{\text{eq}}^{\text{max}} = \frac{L - N_\ell s_{\text{min}}}{N_\ell - 1},$$

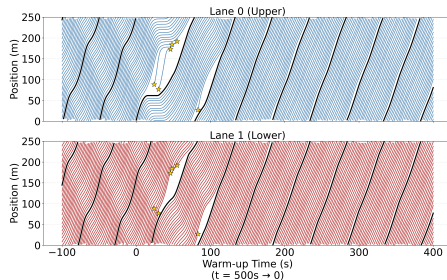
$$s_{\text{eq}}^{\text{max}} = (s_0 + v^* \tau) \left[1 - \left(\frac{v^*}{v_0} \right)^\gamma \right]^{-1/2}.$$

- ▶ Road geometry determines feasible equilibrium spacing.
- ▶ The IDM relation maps spacing to the shared target speed v^* .
- ▶ Both AVs track the same v^* after pairing.

The cruise target respects the density of the effective lane, not only the road speed limit.

The validation isolates the value of each methodological step

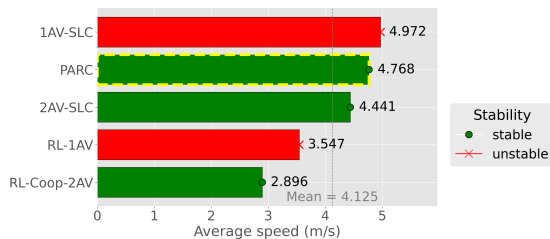
Method	Scientific purpose
AII-HDV	Demonstrate uncontrolled waves
RL-1AV	Test whether one learned AV is sufficient
RL-Coop-2AV	Test whether shared information creates useful coordination
2AV-SLC	Control for AV count without cross-lane coordination
PARC	Test the distilled pair-aligned mechanism



After alignment: nearly parallel trajectories

Source: Paper: Discussion and Comparison; trajectory_log_paired_controller.png

RL identifies the structure; PARC executes it efficiently



$$4.441 \text{ m/s} \rightarrow 4.768 \text{ m/s}$$

$$\frac{4.768 - 4.441}{4.441} \times 100\% = 7.4\%$$

1. One AV cannot reject lane-changing disturbances across both lanes.
2. Cooperative reward design identifies cross-lane alignment as a useful structure.
3. PARC implements that structure transparently and improves stabilized speed.

Current evidence is limited to centralized information, two lanes, two AVs, and simulation-based validation.

Source: Paper: Discussion and Conclusion; avg_speed_comparison.png; reviewer comments



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